

Date: 2026-05-19
Request ID: 100048569

Microwave				
Element(s)	Cu			
SOP#				
Acid Cocktail	3 mL HCl, 1 mL HNO3			
Rack	20			
Vessel	Quartz			
Stir	0%			
Time (min)	Power (W)	T1 (°C)	T2 (°C)	P (bar)
15	1200	180	65	110
15	1500	240	65	110
30	1500	240	65	110
sample(s)	weight (g)	volume (mL)		
200128035_1	0.1039	250		
200128035_2	0.1075	250		
200128036_1	0.1064	250		
200128036_2	0.1064	250		
200128037_1	0.1038	250		
200128037_2	0.1141	250		
200128038_1	0.1112	250		
200128038_2	0.1077	250		

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Cu ICP analysis

sample	Dilution	conc. [mg/L]	Cu_ppm	%Cu
200128035_1	50/6	1.934	38779.27	3.88
200128035_2	50/6	2.072	40155.04	4.02

cu result: 39467.16 ppm

cu result: 3.95 %

cu oxide factor: 1.252

cu oxide result: 4.94 %

cu lot: *Manufacturer: Assurance, 28-83CUT exp: 2026-09-30*

Note(s):

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Cu ICP analysis

sample	Dilution	conc. [mg/L]	Cu_ppm	%Cu
200128036_1	50/4	1.745	51251.17	5.13
200128036_2	50/4	1.765	51838.58	5.18

cu result: 51544.88 ppm

cu result: 5.15 %

cu oxide factor: 1.252

cu oxide result: 6.45 %

cu lot: *Manufacturer: Assurance, 28-83CUT exp: 2026-09-30*

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Cu ICP analysis

sample	Dilution	conc. [mg/L]	Cu_ppm	%Cu
200128037_1	50/4	2.119	63794.56	6.38
200128037_2	50/4	2.339	64061.13	6.41

cu result: 63927.84 ppm

cu result: 6.39 %

cu oxide factor: 1.252

cu oxide result: 8.00 %

cu lot: *Manufacturer: Assurance, 28-83CUT exp: 2026-09-30*

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Cu ICP analysis

sample	Dilution	conc. [mg/L]	Cu_ppm	%Cu
200128038_1	50/3	2.077	77825.24	7.78
200128038_2	50/3	2.018	78071.80	7.81

cu result: 77948.52 ppm

cu result: 7.79 %

cu oxide factor: 1.252

cu oxide result: 9.76 %

cu lot: *Manufacturer: Assurance, 28-83CUT exp: 2026-09-30*

Note(s):

A = crucible

B = crucible + sample

C = crucible + sample after drying

$$\%LOI = ([B-A] - [C-A]) / (B-A) * 100\%$$

$$ppm\ M+ = [conc.][volume][dilution] / [weight]$$

$$\%M+ = [conc.][volume][dilution] / ([weight] * 10,000)$$

$$\%MO = [oxide\ factor] * \%M+$$

$$ppm\ M+ \ [dried] = ppm\ M+ / ([1 - (\%LOI) / 100])$$

$$\%M+ \ [dried] = \%M+ / ([1 - (\%LOI) / 100])$$

$$\%MO \ [dried] = \%MO / ([1 - (\%LOI) / 100])$$

$$\%Cr(VI) = (1.733[mL\ FAS][N\ FAS]) / ([weight])$$

$$\%Cr_2O_3 = 1.462 * [\%Cr(VI)]$$

$$\%Cr(III) = total_Cr - Cr(VI)$$